

Association of *Pota*-phrases with compared phrases in Korean Comparatives

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Jae-Il Yeom. 2016. Association of *Pota*-phrases with compared phrases in Korean comparatives. *Language and Information* 20.2, 55-82. In a comparative sentence in Korean, a *pota*-phrase can occur in various positions in relation to the compared phrase, but both are compared with respect to the same comparative predicate. In this paper, I discuss two major distributional properties of *pota*-phrases. One is that despite the general flexibility of positions, there is a directional asymmetry: a *pota*-phrase can be associated with a topic if it occurs after the topic, but not if it occurs before the topic. The other is that a *pota*-phrase can combine with a much larger phrase than a comparative predicate, even with a syntactic island in between. These properties are explained by analyzing a comparative construction as involving multiple foci, assuming that the comparative morpheme *te* and the compared phrase (or its trace) yield a set of alternatives on the right side of the *pota*-phrase and that a *pota*-phrase is a focus-sensitive operator that exploits the sets of alternatives. In dealing with focus, I adopt a slightly modified version of the structured meaning approach so that the meaning of comparison can be reflected in the alternative set. (Hongik University)

Key words *pota*, comparative, multiple focus, association with compared phrase, structured meaning

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In (2.a) the *pota*-phrase occurs in the relative clause and Inho is compared with a friend of Mina's. This sentence has the corresponding English comparative construction, as given in the meaning of the sentence. In (2.b), on the other hand, Inho is compared with Mina structurally, but actually a friend of Inho's is compared with a friend of Mina's. This does not have a corresponding English sentence. This indicates that Korean comparatives need a different analysis. (2.c) can be interpreted as having the same meaning as (2.a) or (2.b). This means that in Korean, the positions of *pota*-phrases are not fixed, and a different comparative predicate is formed depending on the positions of the *pota*-phrase.

Moreover, if the nominative case marker is replaced with the topic marker, the *pota*-phrase cannot precede the compared phrase:

- (3) a. mina-nun inho-pota cha-lul te cal mo-nun chinkwu-lul manna-ss-ta.
 Mina-nom Inho-than car-acc more well drive-adn friend-acc meet-pst-dec
 b. ??incho-pota mina-nun cha-lul te cal mo-nun chinkwu-lul manna-ss-ta.
 Inho-than Mina-nom car-acc more well drive-adn friend-acc meet-pst-dec

(3b) can be compared with (2.b) in their acceptability. This indicates that topichood is one of the factors that determine the positions of *pota*-phrases.

I briefly mentioned a couple of issues in Korean comparatives, in relation to the positions of *pota*-phrases. In previous analyses of Korean comparatives, these issues are not considered seriously. Kim and Sells (2009) attempt to discuss the distribution of *pota*-phrases, following the idea of Napoli (1983) and Pinkham (1984, 1996), but the data is limited and the observation and the analysis are not satisfactory. And there has been no attempt to discuss the issue of how a *pota*-phrase is associated with the comparative predicate when they are far apart. In this paper, I will discuss more extensive data on the distribution of *pota*-phrases and explain how a comparative construction is interpreted when a *pota*-phrase is far away from a compared predicate, and why *pota*-phrases show such a distribution.

The paper is organized as follows. In Section 2, I briefly introduce properties of Korean comparatives and discuss two distributional restrictions of *pota*-phrases. One is that if a *pota*-phrase follows the topic of the sentence, it can be associated with a compared phrase in any position, even with the topic, but that if a *pota*-phrase precedes the topic of the sentence, it cannot be associated with the topic. And there is a strong tendency that the *pota*-phrase tends to occur before the compared phrase. The other is that a *pota*-phrase is allowed only if there is a comparative predicate, but that the two can occur far from each other, even across a syntactic island. In Section 3, I account for the observations we made in Section 2 by claiming that the comparative morpheme *te* and the compared phrase close to the comparative predicate are focused and generate a set of alternatives. I also claim that a *pota*-phrase is a focus-sensitive

operator that exploits the alternatives. I adopt the structured meaning approach in dealing with focus so that the meaning of comparison can be captured as the condition on the alternative set. In Section 4, I summarize the discussions in the paper.

2. Distributions of *pota*-phrases

2.1 Basic properties of Korean comparatives

Korean comparatives are different from English comparatives in many ways. In Korean, the comparative morpheme *te* is optional in many cases, while in English the comparative morpheme is obligatory.³ And a *pota*-phrase precedes the comparative morpheme, while a *than*-phrase follows the comparative predicate, which is due to the head-initial/head-final parameter among languages. What is more relevant in this paper is that in English, the complement of *than* can be a clause, as in (1.b), while in Korean comparatives, the standard marker *pota* can occur with a NP, PP or AdvP but not with a clause:

- (4) a. Inho-nun Minho-pota (te) ttokttokha-ta.
 Inho-top Minho-than more smart_v-dec⁴.
 'Inho is smarter than Minho. (NP)
- b. Inho-nun ikos-eyse-pota kohyang-eyse te hayngpokha-ta.
 Inho-top here-at-than hometown-at more happy_v-dec
 'Inho is happier at hometown than here.' (PP)

³ I do not know whether there are cases where *te* is obligatory. Kim and Sells (2009) claim that there are, with the following examples as supporting evidence:

- i. phyengso-pota samsip pwun-i *(te) keli-ess-ta
 normal-than 30 minutes-nom *(more) take-pst-dec
 'It took 30 more minutes than usual.'
- ii. nam-pota *(te) mek-ess-ta
 others-than *(more) eat-pst-dec
 '(He) ate more than others.'

But these are not supporting evidence for their claim. In these examples, *te* is obligatory as an argument, not simply as a comparative morpheme: it forms the important part of the issues that the sentences intend to convey.

⁴ In this paper, a stative verb in Korean is often glossed with an English adjective with a subscript *v*.

- c. Inho-nun ppali-pota mwusahi cip-ey ka-ko.sip-ess-ta.⁵
 Inho-top fast-than safely home-to go-want-pst-dec
 'Inho more wanted to go home safely than fast.' (AdvP)
- d. Inho-nun Minho-ka ttokttokha-[*ta, n kes/cengto]-pota (te) ttokttokha-ta.
 Inho-top Minho-nom smart_v-{dec, adn thing/degree}-than more smart_v-dec
 'Inho is smarter than Minho is smart.' (clause)

As (4.d) shows, *pota* cannot take a clause ending with a declarative mood marker. Instead, it can take a *kes*-phrase, which consists of an adnominal clause and a bound noun *kes*. A *kes*-phrase is a NP. Jhang (2001), Park (2008), among others, claim that Korean also has clausal comparatives as well as phrasal comparatives, assuming that *kes* is a complementizer. But *kes* is always preceded by an adnominal expression. It can be pluralized, just like other common nouns, as shown in Yeom (2014). Even if there are cases where it cannot be pluralized, it is implausible to assume that *kes* is ambiguous. In relation to comparatives, *kes* can be replaced with a common noun *cengto* 'degree', when the *kes*-phrase denotes a degree, as shown in (4.d). Thus I will assume that *kes* is a bound noun.⁶

The reason that *pota* only occurs with a NP, PP, or AdvP is that it is a postposition. In Korean, other postpositions show the same properties:

5 In this example, the use of the *pota*-phrase is licensed by the gradable predicate *ko.sip*. Without it, the sentence becomes odd:

- i. ??Inho-nun ppali-pota mwusahi cip-ey ka-ss-ta.
 Inho-top fast-than safely home-to go-pst-dec
 'Inho went home safely than fast.'

Without a gradable predicate, *pota* can be used metalinguistically:

- ii. ??Inho-nun ppali-la-ki-pota mwusahi cip-ey ka-ss-ta.
 Inho-top fast-be.dec-nml-than safely home-to go-pst-dec
 'Inho went home safely rather than fast.'

6 In surface forms, Japanese seems to have comparative constructions in which *yor*i 'than' is preceded by a clause. But Beck, Oda, and Sugisaki (2004) and Sudo (2015) show that the clause is actually an adnominal clause that modifies a null head noun. For this reason, they assume that the clause modifies a null noun and that *yor*i is always preceded by a (noun) phrase. This idea can be extended to English comparatives because the complement clause of *than* is generally analysed as a free relative clause which denotes a degree. We can assume that a degree-denoting free relative clause can be assumed to be a NP, just like a free relative clause headed by *what*, as in What I need is a nice car.

- (5) a. na-nun chayk-(*ul)-pwuthe sey-ess-ta. (NP)
 I-top book-(acc)-from count-pst-dec
 'I counted books first.'
- b. na-nun chayk-ul mith-eyse-pwuthe sey-ess-ta. (PP)
 I-top book-acc bottom-at-from count-pst-dec
 'I counted the books from the bottom.'
- c. na-nun kukes-ul wancenhi-kkaci-nun ani-ciman cal ihayhay-ss-ta.⁷ (AdvP)
 I-top it-acc perfectly-to-top not.be-but well understand-pst-dec
 'I understood it well, even if not perfectly.'

These examples show that a postposition can occur with a NP without a case-marker, a PP and even an AdvP. By assuming that *pota* is a postposition, we can explain why no case-marker is allowed in a *pota*-phrase:

- (6) na-nun Inho-(*lul)-pota Cinho-lul cohaha-n-ta.
 I-top Inho-(acc)-than Cinho-acc like-impf-dec
 'I like Cinho more than Inho.'

Since no case-marker is attached, a sentence can be ambiguous, as observed by Jhang (2001):

- (7) na-nun Inho-pota Cinho-lul te cohaha-n-ta.
 I-top Inho-than Cinho-acc more like-impf-dec
 'I like Cinho more than Inho does.' (Reading 1)
 'I like Cinho more than I do Inho.' (Reading 2)

In (7), the *pota*-phrase can be understood as compared with the speaker or with Cinho.

And more than one *pota*-phrase can occur in a comparative construction. In such a case, one restriction is that all *pota*-phrases must occur at one side with respect to the compared phrase:

- (8) a. Inho-ka Mina-pota Cina-pota te pappu-ta.
 Inho-nom Mina-than Cina-than more busy_v-dec
 'Inho is busier than Mina than Cina.' (Lit.)
 b. ?*Mina-pota Inho-ka Cina-pota te pappu-ta.
 c. Mina-pota Cina-pota Inho-ka te pappu-ta.

⁷ Not all postpositions can occur with an AdvP. *kkaykkushi-pwuthe* 'from clearly' is possible, but *kkaykkushi-eyse* 'at/from clearly' is not possible. I do not know whether it is just a semantic restriction. We need to look into this issue more closely, but it is not within the scope of this paper.

I suppose that two or more *pota*-phrases are conjoined with an implicit conjunctive connective.

- (9) a. Inho-ka [Mina-pota (kuliko) Cina-pota] te pappu-ta.
 Inho-nom Mina-than (and) Cina-than more busy_v-dec
 'Inho is busier than Mina and than Cina.'
 b. [Mina-pota (kuliko) Cina-pota] Inho-ka te pappu-ta.

Then it is naturally explained why one of the conjuncts cannot move out of the whole coordination structure. (Cf. Coordinate Structure Constraint)

In English, if *than* combines with a clause, the clause always denotes a degree. In Korean, a *kes*-phrase can denote a degree, as in (4.d), or an object, as in (10):⁸

- (10) i chayk-i inho-ka sa-n kes-pota te caymiiss-ta.
 this book-nom Inho-nom buy-adn thing-than more interesting_v-dec
 'This book is more interesting than what Inho bought.'

In this example, the *kes*-phrase denotes something like a book. This can be considered a case where *pota* takes an individual-denoting NP, as in (4.a).

This implies that it is not an important factor whether the complement has a clause in it or not. It is more important that the complement of *pota* can denote a degree or a non-degree object. If a *kes*-phrase with a clause denotes a degree, it is expected that there is a corresponding case where the complement of *pota* is a degree-denoting NP. In English, *than* can take a measure phrase:

- (11) John is taller than 6 feet.
 john $\in \{x \mid \exists d[x \text{ is } d\text{-tall} \ \& \ d > 6 \text{ feet}]\}$

The corresponding Korean sentence is a little odd:

- (12) ??John-i 6 ft-pota te khu-ta.
 John-nom 6 ft-than more tall_v-dec
 'John is taller than 6 ft.'

But there are examples in which *pota* can take a measure phrase:⁹

⁸ In Korean a *kes*-phrase can denote other things such as events, facts and propositions, as discussed in Yeom (2015). I will not cover all of them, but such *kes*-phrases also can be used as the standard of comparison.

⁹ There is a restriction on predicates that allow a measure phrase, but it is beyond the scope of this paper.

- (13) *kenmwul-i 3o-pota te kiwul-ess-ta.*
 building-nom 3o-than more tilt-prf-dec
 'The building has tilted by more than 3o.'

Though the use of a measure phrase is more restricted in Korean, it is not impossible. Thus the complement of *pota* can denote anything that the complement of *than* can in English, though *pota* only takes a NP categorically:

(14)

		semantic distinction	
		degree-denoting	individual-denoting
structural distinction	NP with no clause	measure phrase	individual-denoting NP
	<i>kes</i> -phrase	degree-denoting clause	individual-denoting clause

This implies that structural differences between Korean and English are not important.

2.2. Positions of *pota*-phrases in relation to compared phrases

In this section, I will consider the distributions of *pota*-phrases. First, I will show where a *pota*-phrase occurs in relation to other phrases, including compared phrases. One restriction in the position of a *pota*-phrase is that a *pota*-phrase must occur to the left side of the comparative predicate. The following sentences are odd because they do not satisfy the condition:¹⁰

¹⁰ There are cases where it might seem that a gradable predicate precedes a *pota*-phrase:

- i. *te manhun salam-tul-i pay-pota sakwa-lul mek-ess-ta.*
 more many people-pl-nom pear-than apple-acc eat-pst-dec
 'More people ate apples than pears.'

But this is not a counterexample to the condition that a *pota*-phrase precedes a comparative predicate. In this case no gradable predicate is necessary:

- ii. *salam-tul-i pay-pota sakwa-lul mek-ess-ta.*
 people-pl-nom pear-than apple-acc eat-pst-dec
 'People ate apples rather than pears.'

In this example, *pota* means 'rather than', which does not require a gradable predicate.

- (15) a. *Inho-ka cha-lul te Cinho-pota cal mo-n-ta.
 Inho-nom car-acc more Cinho-than well drive-impf-dec
 'Inho drives better than Cinho.'
 b. *Inho-ka te coh-un cha-lul Cinho-pota kaci-ko.iss-ta.
 Inho-nom more good-adn car-acc Cinho-than have-impf-dec
 'Inho has a better car than Inho.'

As far as a *pota*-phrase occurs to the left of the comparative predicate, its position is not fixed:

- (16) a. Inho-ka cha-lul Cinho-pota te cal mo-n-ta.
 Inho-nom car-acc Cinho-than more well drive-impf-dec
 'Inho drives a car better than Cinho.'
 b. Inho-ka Cinho-pota cha-lul te cal mo-n-ta.
 d. Cinho-pota Inho-ka cha-lul te cal mo-n-ta.

Despite the variability of the positions of *pota*-phrases, there are some tendencies related to information structure. First, it is more natural to put a *pota*-phrase before the compared phrase. The tendency is more salient when the phrases involved are PPs, and it is stronger when the PPs are arguments. If a PP is an argument in a VP, it is odd to put the compared phrase before the *pota*-phrase:

- (17) a. Inho-nun hakkyo-ey-pota cip-ey mence ka-ss-ta.
 Inho-top school-to-than home-to earlier go-pst-dec
 'Inho went home earlier than to school.'
 b. hakkyo-ey-pota Inho-nun cip-ey mence ka-ss-ta.
 c. ??Inho-nun cip-ey hakkyo-ey-pota mence ka-ss-ta.
 d. ?*cip-ey inho-nun hakkyo-ey-pota mence ka-ss-ta.

This clearly shows that the default order is to put a *pota*-phrase before the compared phrase.

The same default ordering is observed even if a PP is not an argument:

- (18) a. Inho-nun silnay-eyse-pota yaoy-eyse te ppalli ttwuy-n-ta.
 Inho-top indoor-at-than outdoor-at more fast run-impf-dec
 'Inho runs faster indoors than outdoors.'
 b. ??Inho-nun yaoy-eyse silnay-eyse-pota te ppalli ttwuy-n-ta.
 c. ??yaoy-eyse Inho-nun silnay-eyse-pota te ppalli ttwuy-n-ta.

In cases where the predicate is a stative intransitive verb, the reversed ordering is sometimes acceptable, but only if the compared phrase is marked with a nominative case-marker:¹¹

- (19) a. *cenca-uy wuntonglyang-uy pwulhwakcengseng-un A-eyse-pota B-eyse*
 electron-of quantity.of.motion-of uncertainty-top A-at-than B-at
 khu-ta.
 large-dec
 'The uncertainty in the quantity of motion of an electron is larger in B than in A.'
 b. *cenca-uy wuntonglyang-uy pwulhwakcengseng-un B-eyse-?(ka)*
 electron-of quantity.of.motion-of uncertainty-top B-at-(nom)
 *A-eyse-pota khu-ta.*¹²
 A-at-than large-dec

In (19.b), the nominative case-marker is required and the phrase carries a strong focus.

A similar observation is made when the compared phrase is the object of a sentence, though it is not as salient:

- (20) a. *Inho-nun yenkuk-pota yenghwa-(lul) (te) cohaha-y.*
 Inho-top play-than movie-acc more like-dec
 'Inho likes movies more than plays.'
 b. *Inho-nun yenghwa-?(lul) yenkuk-pota (te) cohaha-y.*

Even if the ordering of a *pota*-phrase and its compared phrase is flexible, (20.a) is more natural than (20.b). In a colloquial style, an accusative case-marker is often omitted, as in (20.a). But it is not the case in (20.b). This again indicates that when the compared phrase precedes the *pota*-phrase, it must be marked with an accusative case-marker. We can assume that the case-marker plays the role of marking focus.

In discussing these observations, we need to consider the fact that a comparative construction involves focus. If two things are compared, the corresponding phrases are supposed to be focused. But they are not evenly focused. The primary focus tends to go to the compared phrase. This can be supported by the fact that in a pseudocleft it is the compared phrase, but not the *pota*-phrase, that is focused:

11 But a case-marker *-ka* is not always used to express focus. It does not seem that in a sentence with a transitive verb, *-ka* occurs with a PP:

i. *Inho-nun ton-ul Busan-eyse-?(ka) Seoul-eyse-pota te pel-ess-ta.*

ka cannot be added to the PP even if the compared phrase precedes the *pota*-phrase. I do not know if there is any previous research on the nominative case marker attached to a PP.

12 A similar sentence is used in Korean SAT.

- (21) a. Inho-ka yenkuk-pota (te) cohaha-n kes-un yenghwa-i-ta.
 Inho-nom play-than more like-adn thing-top movie-be-dec
 'It is movies that Inho likes more than plays.'
 b. ?*Inho-ka yenghwa-lul (te) cohaha-n kes-un yenkuk-pota-i-ta.
 Inho-nom movie-acc more like-adn thing-top play-than-be-dec
 'It is than plays that Inho likes movies more.'

The compared phrase can be located in the precopular position, but the *pota*-phrase cannot. This indicates that the target of comparison is part of the main assertion, which makes it possible to emphasize it. But the standard of comparison is part of the background for the assertion, which makes it odd to place it at the focus position, separately from the assertion.

This does not mean that a *pota*-phrase can never be focused. In a pseudocleft, the *kes*-phrase presupposes the meaning of uniqueness or exhaustivity. That is, the precopular phrase must be the only thing that can fill a gap in the *kes*-phrase.¹³ (21.b) is odd because no exhaustivity is implicated by the focused *pota*-phrase. When no uniqueness is required, a *pota*-phrase can be focused, preferably with the delimiter *-to* 'also, even'. Consider (20) again, with the addition of *-to* 'also, even':

- (22) a. Inho-nun yenkuk-pota-(to) yenghwa-lul (te) cohaha-n-ta.
 Inho-top play-than-also movie-acc more like-dec
 'Inho likes movies more than plays.'
 b. Inho-nun yenghwa-lul [yenkuk-pota]_F-(to) (te) cohaha-n-ta.
 Inho-top movie-acc play-than-also more like-dec
 'Inho likes movies more than even plays.'

In (22.a), if *-to* is added to the *pota*-phrase and the phrase precedes the compared phrase, it preferably means 'rather than', with no emphasizing effect. It does not tend to yield a set of alternatives to plays that are compared with movies. It seems that we are only considering the two alternatives, movies and plays. This again shows that a *pota*-phrase does not tend to be focused, even with the marking of *-to*. In (22.b), on the other hand, the use of *-to* yields a set of other alternatives that Inho prefers movies to. By placing the *pota*-phrase just before the comparative predicate, it is really focused and has the effect of enhancing the degree of liking movies by comparing them with plays, with the help of the implication that Inho likes plays very much. This comes from the scalar interpretation of *-to*.

To capture the observations we have made so far, we can make the following assumptions:

¹³ This implies that the focus is identificational in the sense of Kiss (1998).

- (23)
- a. In the base-generated structure, a *pota*-phrase precedes the compared phrase.
 - b. In the base-generated structure, the compared phrase is focused.
 - c. If the compared phrase moves to the left of the *pota*-phrase, it should be marked with a case-marker, which plays the role of focus marking.

In the base-generated structure, a *pota*-phrase precedes the compared phrase, which makes the compared phrase close to the verb. This reflects the tendency that the compared phrase is focused. But if the compared phrase is scrambled over the *pota*-phrase, the focus on the compared phrase is marked with a case-marker, instead, if a case-marker is allowed. On the other hand, a nominative case-marker cannot save (17.c), (17.d) and (18.b-c), because the predicates in them do not license a nominative case-marker that occurs with a PP.

One exception to the generalizations in (23.c) is a case where the compared phrase is a topic. If the compared phrases are topicalized, the corresponding *pota*-phrases cannot precede them:

- (24)
- a. Inho-nun Cinho-pota te coh-un cha-lul kaci-ko.iss-ta.
Inho-top Cinho-than more good_v-adn car-acc have-impf-dec
'Inho have a better car than Cinho.'
 - b. ?*Cinho-pota Inho-nun te coh-un cha-lul kaci-ko.iss-ta.

In (24.a), we saw that the preceding compared phrase can be the topic of the sentence. This can be compared with (24.b), where the compared phrase with a topic marker follows the *pota*-phrase, which makes the sentence odd. This observation can be generalized as follows:

- (25) A *pota*-phrase can follow any compared phrase, but can precede only a non-topic compared phrase.

We need to find a way that these observations fit with the condition that a compared phrase is focused. One way is to assume the following:

- (26) A *pota*-phrase is associated with the compared phrase, or a trace of its, to its right.

Together with (23.b), the compared phrase or its trace that follows the *pota*-phrase is focused. Under the assumption that a topic is not focused, this allows us to explain why in (24.b), the *pota*-phrase is not associated with the topic. In (24.a), on the other hand, the topic precedes the *pota*-phrase, which means that there is a trace of the topic that follows the *pota*-phrase. In this case, we can assume that the *pota*-phrase is associated with the trace.

One question is whether a null element like a trace can be focused. And the answer is positive:

- (27) na-nun Inho-ka ceyil coh-a. nay-ka tto e_F cohaha-nun salam-un Mina-i-a.
 I-top Inho-nom most like-dec I-nom also like-adn person-top Mina-be-dec
 'I like Inho most. A person I also like e_F is Mina.'

The focus-sensitive operator *tto* 'also' is associated with the null element e_F, which needs to yield a set of alternatives. Thus there is nothing strange in assuming that a *pota*-phrase is associated with the trace of the compared phrase.

2.3. Positions of *pota*-phrases in relation to gradable predicates

In this section, I will show that a *pota*-phrase can be far away from the gradable predicate. A *pota*-phrase occurs with a gradable predicate because a comparative construction conveys the meaning of comparison of degrees. If there is no gradable predicate, the sentence becomes odd:

- (28) ??ikes-un ceket-pota te sasil-i-ta.
 this-top that-than more fact-be-dec
 'This is more a fact than that.'

This sentence is odd because *sasil-i* 'be true, be a fact' is not gradable. A non-gradable predicate does not allow an expression like *ceket-pota te* '-er than that'.

Considering the dependency of a *pota*-phrase on a gradable predicate, the *pota*-phrase in (29) is dependent on the use of *khu* 'be big':

- (29) a. ?*Inho-nun Mina-pota sakwa-lul mek-ess-ta.
 Inho-top Mina-than apple-acc eat-pst-dec
 'Inho ate an apple than Mina.'
 b. Mina-pota Inho-ka te khu-n sakwa-lul mek-ess-ta.
 Mina-than Inho-nom more big_V-adn apple-acc eat-pst-dec

In (29.a), the use of the *pota*-phrase is not acceptable because there is no gradable predicate that can license the *pota*-phrase. In (29.b), the sentence is saved by the use of *te khun* 'more big'. It is a comparative predicate, that is, a gradable predicate with the comparative morpheme, that licenses the use of the *pota*-phrase.

This might lead to the possibility that the *pota*-phrase forms a constituent with the comparative predicate. But if this were the case, *Mina-pota te khun* might mean 'bigger than Mina' but this is not the correct reading. The other possibility is that the phrase means 'bigger than an/the apple that Mina ate', assuming *Mary* means 'an/the apple

that Mina ate'. This is not structurally supported:

- (30) A: Inho-nun **elmana** **khu-n** sakwa-lul mek-ess-nya?
 Inho-top how big_v-adn apple-acc eat-pst-int
 'How big an apple did Inho eat?'
 B: ??(Inho-nun) [Mina-pota te khu]-n sakwa-lul mek-ess-e.
 Inho-top Mina-than more big_v-adn apple-acc eat-pst-dec
 'Inho ate an apple bigger than (the/an apple) Mina (ate).'

The question with *elmana khu-n* 'how big' is answered with the phrase *Mina-pota te khu-n* 'bigger than (an apple that) Mina (ate)' and the sentence becomes odd. This shows that *Mina-pota te khu-n* cannot form a constituent, regardless of how the phrase is interpreted. This also means that *Mina* does not refer to the apple that Mina ate.

A *pota*-phrase can be attached to a much more complex predicate:

- (31) Inho-nun Mina-pota te olay-toy-n kose-lul chac-un hakca-lul
 Inho-top Mina-than more old-become-adn old.book-acc find-adn scholar-acc
 chwuycayhay-ss-ta.
 interview-pst-dec
 'Inho interviewed a scholar who had found an older book than Mina.' (Lit.)

In (31), Mina is compared with Inho with respect to the property of being old of the books which had been found by two scholars who they interviewed. The phrases that denote the comparison standard and comparison target occur in the matrix clause, but the comparative predicate *te olay-toy* 'become older' is deeply embedded in the relative clause. The *pota*-phrase cannot be interpreted directly with the comparative predicate because the sentence does not mean the relevant book is older than Mina.

I dealt with examples in which NPs are compared. We can make the same observations in examples in which PPs are compared:

- (32) Inho-nun Busan-eyse-pota Seoul-eyse [NP[CP [NP[CP te olaytoy-n] chayk-ul]
 Inho-top Busan-in-than Seoul-in more old_v-adn book-acc
 kaci-ko.iss-nun] salam-ul] manna-ss-ta.
 have-impf-adn person-acc meet-pst-dec
 'Inho met a person who has an older book in Seoul than in Pusan.'

A book is not older in Seoul than in Busan. The locative PPs are interpreted in relation to the event of meeting. Thus the *pota*-phrase does not form a constituent with the comparative predicate. The *pota*-phrase is associated with the comparative predicate across two Complex NPs.

This is not a result of scrambling. Scrambling over a Complex NP is not allowed:

- (33) *[Mina-eykey]_i Inho-nun nay-ka t_i pha-n cha-lul kaci-ko.iss-ta.
 Mina-to Inho-top I-nom sell-adn car-acc have-impf-dec
 'Inho is keeping a car I am going to sell to Mina.'

Mina-eykey moved out of the relative clause, which makes the sentence odd. Thus a *pota*-phrase associated with a comparative predicate across a syntactic island is not base-generated within the syntactic island and scrambled to the matrix clause. Instead the *pota*-phrase is base-generated in the matrix clause. The *pota*-phrase and the comparative predicate never form a constituent in any syntactic process.

(29.b) roughly means that the apple that Inho ate is bigger than the one that Mina ate. That is, two apples are compared. But the sentence is structurally expressed as if two persons were compared. The question is how the comparison of two persons is converted to the comparison of the sizes of the apples that they ate. To get the right meaning, we might assume that *te* is attached to a larger phrase *khu-n sakwa-lul mek* 'eat a big apple'. But this is not plausible. Semantically, *te* is licensed by the gradable predicate *khu* 'be big'. Moreover, *te* cannot move away from the comparative predicate:

- (34) ??Mina-pota *te* Inho-ka khu-n sakwa-lul mek-ess-ta.
 Mina-than more Inho-nom big_v-adn apple-acc eat-pst-dec

Thus we conclude that the *pota*-phrase belongs to the main clause, while *te* belongs to the clause formed by the comparative predicate.

So far I have shown that when non-degree NPs or PP are compared, they can be from from the comparative predicate. On the other hand, a degree-denoting NP or degree-denoting *kes*-phrase cannot occur far from the gradable predicate:

- (35) a. ?*3o-pota Inho-nun *te* kiwu(l)-n kenmwul-ul palkyenhay-ss-ta.
 -than Inho-top more tilt-adn building-acc find-pst-dec
 'By more than 3o Inho found a building that tilted.' (Lit.)
 b. ?*Mina-ka ttoktokha-n kes-pota Inho-nun *te* ttoktokha-n salam-ul
 Mina-nom smart_v-adn thing-than Inho-top more smart_v-adn person-acc
 manna-ss-ta.
 meet-pst-dec
 'Than Mina was (smart) I met a person who was smarter.' (Lit.)

In these examples, where the *pota*-phrases occur in the matrix clauses, while the gradable predicates are in the relative clauses. This makes the sentences odd. This indicates that *pota*-phrases behave differently depending on whether the complement of *pota* denotes a degree or a non-degree semantic entity. A structural difference is not a factor for the difference.

An analysis of comparatives in Korean must be able to interpret comparative sentences with a *pota*-phrase in various positions in a consistent way. In the next section, I will propose the analysis which allows us to do so.

3. Interpretation of comparatives

3.1. Interpretation of a complex comparative predicate

In English, *than* can be followed by a clause or a non-clausal phrase. In Korean, *pota* is not preceded by a clause, assuming that *kes* is a bound noun.¹⁴ Then it may be right to claim that Korean has only phrasal comparatives, as claimed in Kim and Sells (2010).

The contrast between Korean comparatives and English comparatives is not clear as it seems. In a clausal comparative in English, the complement of *than* is analyzed as a free relative headed by a null operator that ranges over degrees. Thus the meaning of (1.b) is like the following:

- (36) John is taller than [_{CP} Op_i [Bill is e_i (tall)]]
 $\exists d[\text{tall}(d)(\text{john}) \ \& \ d > \max(\{d' \mid \text{tall}(d')(\text{bill})\})]$

Von Stechow (1984) assumes that the complement of *than* refers to the maximal degree of Bill's height. A free relative that denotes a degree can be considered a NP, just like a free relative clause with *what*. And *than* can in fact take a degree-denoting NP, as shown in (11). Thus there is no big difference between (1.b) and (11). Furthermore, *pota* is not quite different from *than* in that it takes a degree-denoting NP, as in (13), or in the form of a *kes*-phrase.

In Korean, *pota* can also take an individual-denoting NP or a location-denoting PP as an argument, just as *than* does. Basically a comparative constructure compares degrees. Thus we need to get degrees from individuals or locations in examples like (4.a) and (10.a), respectively. If the complement of *pota* is not a degree-denoting NP, previous analyses propose that the gradable predicate apply to the complement of *pota*, as well as to the compared phrase. For example, to get the interpretation of (4.a), the meaning of the comparative morpheme *-er* in English or *te* in Korean could be defined as follows, following Kennedy (1997):

14 The P *pota* can be preceded by a *kes*-phrase.

I. na-nun [Mary-ka [e] ssun kes]-pota te kin nonmwun-ul sse-ss-ta.
 I-top Mary-nom wrote comp-than more long paper-acc write-past-dec
 'I wrote a longer paper than Mary did.'

There are some linguists like Lee (2002), and Park (2009) that claim that *kes* is a complementizer that heads a CP. They would claim that *pota* is used with a clause.

- (37) $\llbracket te \rrbracket = \lambda G \lambda x \lambda y \exists d [G(d)(y) \ \& \ d > \max(\{d' \mid G(d')(x)\})]$, where
 G: the meaning of a gradable predicate;
 x: the standard of comparison
 y: the target of comparison
- (38) $\llbracket Inho-nun \ Minh-pota \ (te) \ ttokttokha-ta \rrbracket$
 $= \llbracket te \rrbracket (\llbracket ttokttokha \rrbracket) (\llbracket Minh \rrbracket) (\llbracket Inho \rrbracket)$
 $= \exists d [\text{smart}(d)(inho) \ \& \ d > \max(\{d' \mid \text{smart}(d')(minho)\})]$

In the definition of the meaning of *te* given in (37), *G* applies twice. Considering the fact that *G* applies once if the complement of *pota* is a degree-denoting expression, as in (13), the twofold application of *G* must be further motivated.

Moreover, to make this interpretation in (38) work in general, *te* has to be interpreted with *pota* so that we can get the degree of smartness of the complement of *pota*. In Korean, however, this way of interpretation does not work, as shown in (29.b). If the same way of interpretation applied to (29.b), we would get the following meaning:

- (39) $\llbracket te \ khu \rrbracket = \lambda y \lambda x \exists d [\text{large}(d)(x) \ \& \ d > \max(\{d' \mid \text{large}(d')(y)\})]$
 $\llbracket Mina-pota \ te \ khu \rrbracket = \lambda x \exists d [\text{large}(d)(x) \ \& \ d > \max(\{d' \mid \text{large}(d')(mina)\})]$
 $\llbracket Mina-pota \ te \ khu-n \ sakwa \rrbracket$
 $= \lambda x \exists d [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ d > \max(\{d' \mid \text{large}(d')(mina)\})]$

This denotes the meaning 'apples that are larger than Mina'. This interpretation is not acceptable. We also saw in (34) that *te* cannot combine with a larger constituent.¹⁵

This shows that (37) is not the right interpretation. In (29.b), *te* is interpreted when the embedded relative clause is interpreted, regardless of where the *pota*-phrase is, and the *pota*-phrase must be interpreted when the main clause is interpreted. The next question is how the comparative morpheme *te* is associated with the *pota*-phrase, when they are far apart. This cannot be done by a syntactic movement, because the association should be made over a syntactic island, as we saw in (29.b) and (32). One alternative is the mechanism of syntactic binding. But there is no phrase that can be co-indexed with *te*.

Instead, we can think of the association of a *pota*-phrase with *te* as an association with a focus phrase. In association with a focus phrase, if there is a definite or specific NP in between, the association is not possible:

¹⁵ One way out might be to assume that the phrasal complement of *pota* can denote a degree, just as a clausal comparative does, as in English. That is, we might assume that *Mina* denotes the size of the apple that Mina ate. Then *Mina-pota te khuun* would be a constituent. But we saw in (30) that it is not a constituent. This shows that the phrasal complement of *pota* does not denote a degree. Thus a phrasal comparative is different from a clausal comparative, as claimed by Hankamer (1973), Hoeksema (1983), Pinkal (1990), Kennedy (1999), Pancheva (2006).

- (40) Inho-nun [Mina-ka]_F manna-n ({??ku, ??etten}) salam-man chotayhay-ss-ta.
 Inho-top Mina-nom meet-adn ({the, a.certain}) person-only invite-pst-dec
 'Inho invited only {a, ??the, ??a certain} person who Mina_F met.'

The delimiter *-man* is a focus-sensitive operator, and *Mina-ka* is focused. For the proper interpretation of the focus-sensitive operator, the alternatives generated by the focused phrase must be maintained up to the operator. If the definite or specific determiner occurs in between, the reference of the definite/specific NP is fixed and the variability of alternatives is not maintained. This makes the sentence odd.

In interpreting a comparative construction with an individual-denoting NP, two individuals are compared with respect to the property denoted by one comparative predicate. Thus we are concerned with two sets of alternatives: two sizes of apples and two individuals who ate an apple of one size of the two:

- (41) $\exists x \exists y \exists d [\text{apple}(x) \ \& \ \text{eat}(x)(\text{inho}) \ \& \ \text{apple}(y) \ \& \ \text{eat}(y)(\text{mina}) \ \& \ \text{large}(d)(x) \ \& \ d > \max(\{d' \mid \text{large}(d')(y)\})]$

The variables x and y are individual alternatives, and d and d' are degree alternatives. In order for two individuals to be compared in terms of two degrees for a property, the variability of degrees must be maintained from the meaning of the compared predicate up to the point where the two individual-denoting phrases are interpreted. One test for whether alternatives are involved is to see whether a definite or a specific indefinite NP is allowed or not, because they prohibit alternative degrees from being propagated:

- (42) ??Inho-nun Mina-pota te khu-n {ku, etten} sakwa-lul mek-ess-ta.
 Inho-top Mina-than more big_v-adn {the, a certain} apple.acc eat-pst-dec
 'Inho ate {a, ??the, ??a certain} bigger apple than Mina.'

ku makes the NP definite, and *etten* makes it specific. Neither is used in the comparative. If there is a node which has a fixed reference, the variability that arises within the node has no semantic effect over the node. And the sentence becomes odd.

To capture the variability of individuals and degrees, we need to assume that there are multiple foci involved: one is on *te* and the other is on the compared phrase (or a trace of its). We saw that a compared phrase tends to be focused, so it is plausible to assume that a compared phrase generates a set of alternatives. The comparative morpheme *te* also needs an alternative because we cannot use the morpheme when we have only one degree. Thus the idea that the two sets of alternatives are involved in a comparative construction is well motivated. The two sets of alternatives must be exploited by a focus-sensitive operator. I assume that a *pota*-phrase is the focus-sensitive operator.

In dealing with a focus construction, I will resort to the structured meaning approach, following Krifka (1992). The reason is that the structured meaning approach can be more flexible, and it is possible to modify the structured meaning so that the meaning of comparison can be carried as a condition on the alternative set up to the point where a *pota*-phrase is interpreted. On the other hand, structured meanings cause some complexity in the semantics. Thus I will not cover all the interpretation rules necessary to get the meaning of a sentence involving multiple foci. See Krifka (1992) for detailed discussions of them. I will discuss them minimally for the purpose of this paper.

First, a focused expression yields a structured meaning consisting of B(ackground), F(ocus), and a set of ALT(ernatives):

- 43) $\llbracket [a]_F \rrbracket = \langle \lambda x[x], \llbracket a \rrbracket, \text{ALT}(\llbracket a \rrbracket) \rangle$, where
 $\lambda x[x]$: B(ackground);
 $\llbracket a \rrbracket$: F(ocus);
 $\text{ALT}(\llbracket a \rrbracket)$: (a set of) ALT(ernatives) to $\llbracket a \rrbracket$.

Often a structured meaning consists of Background and Focus. But I add a set of alternatives so that I can restrict the set reflecting the meaning of *te*.

A structured meaning combines with the meaning of another expression. In a constituent $[\alpha \beta]$, if α has the structured meaning of the form $\langle \lambda x[B'], F, \text{ALT} \rangle$ type and β does not, it is interpreted as follows:

- 44) $\llbracket \alpha \rrbracket = \langle \lambda x[B'], F, \text{ALT} \rangle$
 $\llbracket [\alpha \beta] \rrbracket =$ i) $\langle \lambda x[B'(\llbracket \beta \rrbracket)], F, \text{ALT} \rangle$, if $\llbracket \beta \rrbracket$ is a semantic argument of B' ;
 ii) $\langle \lambda x[\llbracket \beta \rrbracket(B')], F, \text{ALT} \rangle$, if B' is a semantic argument of $\llbracket \beta \rrbracket$.

Here B' is part of the Background that takes part in the compositional interpretation until the structured meaning meets a focus-sensitive operator. Note that the meaning of β is incorporated within the scope of λx , in which x ranges over alternatives. We have two possible interpretations, depending on the semantic type of β .

If both α and β have the structured meaning, the meaning of $[\alpha \beta]$ is defined in terms of a tuple, in two ways depending on the semantic types of α and β :

- 45) $\llbracket \alpha \rrbracket = \langle \lambda x[B'], F, \text{ALT} \rangle$; $\llbracket \beta \rrbracket = \langle \lambda y[B''], F', \text{ALT}' \rangle$
 $\llbracket [\alpha \beta] \rrbracket =$ i) $\langle \lambda \langle x, y \rangle [B'(B'')], \langle F, F' \rangle, \{ \langle a, b \rangle \mid a \in \text{ALT}, b \in \text{ALT}' \} \rangle$;
 ii) $\langle \lambda \langle x, y \rangle [B''(B')], \langle F, F' \rangle, \{ \langle a, b \rangle \mid a \in \text{ALT}, b \in \text{ALT}' \} \rangle$.

This is the rule to capture multiple foci.

If a set of alternative pairs or alternative n-tuples arises, there must be an operator that exploits the structured meaning. In a comparative construction, I assume that a *pota*-phrase is a focus-sensitive operator. The meaning of *pota* is defined as follows:

$$(46) \quad \llbracket \alpha\text{-pota } \beta \rrbracket = \exists d[B(F) \ \& \ B(\langle d, \llbracket \alpha \rrbracket \rangle)], \text{ where} \\ \llbracket \beta \rrbracket = \langle B, F, ALT \rangle; \langle d, \llbracket \alpha \rrbracket \rangle \in ALT$$

A *pota*-phrase combines with a structured meaning in which a variable ranging over degrees is included. Once a *pota*-phrase is interpreted, the meaning is not structured any more and the interpretation of the rest of a sentence just follows the standard rules of interpretation.

Now I will show how these rules apply in the actual interpretation of the comparative sentence (29.c), the LF of which is given here:

$$(47) \quad \begin{array}{l} [\text{IP} \text{ Mina-pota } [\text{IP} [\text{Inho-ka}]_F [\text{VP} [\text{NP} [\text{CP } e_j [\text{VP}_2 [(\text{te}_i)]_F \text{ khu}]]\text{-}n_i] \text{ sakwa-lul}]] \\ \text{Mina-than} \quad \text{Inho-nom} \quad \text{more} \quad \text{big}_V\text{-adn} \quad \text{apple-acc} \\ \text{mek}]]\text{-ess]-ta.} \\ \text{eat-pst-dec} \end{array}$$

I assume that the standard marker *te* is focused and its interpretation yields a set of alternative degrees.¹⁶ For simplicity's sake, I assume that *te* carries an index the value of which is contextually determined by an assignment, as shown in (48.b).¹⁷ The relation between the two degrees is not directly specified in the interpretation of the comparative predicate. It is captured by restricting the alternative degrees in the structured meaning of *te*. The structured meaning combines with the meaning of the gradable predicate, as in (48.c):

16 I also said that in Korean *te* is optional. We can assume that a null *te* is focused, just as a trace of a compared phrase can be focused.

17 More precisely, we could assume that there are two null elements for degree phrases involved in using a comparative morpheme *te*:

i. Inho-nun khi-ka 170cm-i-ta. kuliko Minho-nun 5cm te e khu-ta.
Inho-top height-nom 170cm-be-dec and Minho-top 5cm more tall_V-dec
'Inho's height is 170cm. And Minho is 5cm taller than Inho.'

In interpreting *te* we need a null element that refers back to 170cm and another degree phrase 5cm.

- (48) a. $\llbracket \text{khu} \rrbracket = \lambda d \lambda x [\text{large}(d)(x)]$
 b. $\llbracket [\text{te}_i]_F \rrbracket^g = \langle \lambda d [\text{d}], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 c. $\llbracket [\text{VP1 } [\text{te}_i]_F \text{ khu}] \rrbracket^g = \langle \lambda d \lambda x [\text{large}(d)(x)], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$

The relation between two degrees later becomes part of the meaning of the sentence when the *pota*-phrase is interpreted.

Then the empty subject is interpreted as in (48.d), but the value of $g(i)$ is abstracted over when the adnominalizer $-n_i$ is interpreted in the relative clause, as in (48.e). (48.f) shows how the resulting meaning is conjoined with the meaning of *sakwa* 'apple'. In (48.g), the NP is used as an argument of the verb *mek* 'eat':

- (48) d. $\llbracket [\text{TP } e_j [\text{te}_i]_F \text{ khu}] \rrbracket^g = \langle \lambda d [\text{large}(d)(g(i))], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 e. $\llbracket [\text{CP } [e_j [\text{te}_i]_F \text{ khu}-n_i] \rrbracket^g = \langle \lambda d \lambda x \llbracket [\text{TP } e_j [\text{te}_i]_F \text{ khu}] \rrbracket^{g(x/i)}, g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 $= \langle \lambda d \lambda x [\text{large}(d)(x)], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 f. $\llbracket [\text{NP } [\text{CP } e_j [\text{te}_i]_F \text{ khu}-n_i] \text{ sakwa}] \rrbracket^g$
 $= \langle \lambda d \lambda P \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ P(x)], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 g. $\llbracket [\text{VP2 } [\text{te}_i]_F \text{ khu}-n \text{ sakwa-lul mek}] \rrbracket^g$
 $= \langle \lambda d \lambda y \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$

The compared phrase *Inho-ka* also yields a set of alternatives. Then the two structured meanings combine with respect to a set of pairs of a degree and an individual, as in (48.i):

- (48) h. $\llbracket \text{Inho-ka}]_F \rrbracket = \langle \lambda y [y], \text{incho}, D_e \rangle$
 i. $\llbracket [\text{Inho-ka}]_F [\text{VP2 } e_j [\text{te}_i]_F \text{ khu}-n_i \text{ sakwa-lul mek}] \rrbracket^g$
 $= \langle \lambda \langle d, y \rangle \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)], \langle g(i), \text{incho} \rangle, \{ \langle d', y \rangle \in D_d \times D_e \mid d' < g(i) \} \rangle$

Now the *pota*-phrase combines with the structured meaning, as given in (48.j), and we get the final derivation of the meaning of the sentence, ignoring the meaning of Tense, as in (48.k):

- (48) j. $\llbracket \text{Mina-pota } \alpha \rrbracket = \exists d'' [B(F) \ \& \ B(\langle d'', \text{mina} \rangle)], \text{ where}$
 $\llbracket \alpha \rrbracket = \langle B, F, \text{ALT} \rangle; \langle d'', \text{mina} \rangle \in \text{ALT}$
 k. $\llbracket \text{Mina-pota } [\text{Inho-ka}]_F [\text{VP2 } [\text{te}_i]_F \text{ khu}-n \text{ sakwa-lul mek}] \rrbracket^g$
 $= \exists d'' [\lambda \langle d, y \rangle \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)] (\langle g(i), \text{incho} \rangle) \ \& \ \lambda \langle d, y \rangle \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)] (\langle d'', \text{mina} \rangle)]$
 $= \exists d'' \exists x [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(\text{incho})] \ \& \ \exists z [\text{apple}(z) \ \& \ \text{large}(d'')(z) \ \& \ \text{eat}(z)(\text{mina})], \text{ where } g(i) > d''$
 $= \exists x \exists z \exists d'' [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(\text{incho}) \ \& \ \text{apple}(z) \ \& \ \text{large}(d'')(z) \ \& \ \text{eat}(z)(\text{mina}) \ \& \ g(i) > d'']$

The background is exploited twice: it combines with both the focus and a pair in the set of alternatives. This yields the meaning of comparison of two individuals and the sizes of apples they ate. Since a *pota*-phrase must combine with a structured meaning involving a set of alternative degrees, *te* must be located in the scope of, thus to the right of, the *pota*-phrase.

3.2. Positions of a compared phrase

In the previous subsection, I have dealt with a case where a *pota*-phrase precedes a compared phrase. To make this analysis work consistently in a case like (29.b), where the compared phrase precedes the *pota*-phrase, we need to assume that there is a focused trace of the compared phrase. In (49), e_k is the trace:

- (49) [TP Inho-nunk [VP Mina-pota [VP [NP e_k]_F [NP [CP e_j [VP₂ [te]_F khu-n] sakwa-lul] mek]]-ess]-ta.

The focused trace yields a set of alternatives, and together with the set of degree alternatives from *te*, the meaning is structured with a set of alternative pairs of a degree and an individual:

- (50) a. $\llbracket [VP_2 [te]_F \text{ khu-n sakwa-lul mek}] \rrbracket$
 $= \langle \lambda d \lambda y \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 b. $\llbracket [e_k]_F \rrbracket = \langle \lambda y [y], g(k), D_e \rangle$
 c. $\llbracket [e_k]_F [VP_2 [te]_F \text{ khu-n sakwa-lul mek}] \rrbracket$
 $= \langle \lambda \langle d, y \rangle \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)],$
 $\langle g(i), g(k) \rangle, \{ \langle d', y \rangle \in D_d \times D_e \mid d' < g(i) \} \rangle$

Then we can interpret the *pota*-phrase:

- (50) d. $\llbracket \text{Mina-pota } [e]_F [VP_2 [te]_F \text{ khu-n sakwa-lul mek}] \rrbracket$
 $= \lambda \langle d, y \rangle \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)] (\langle g(i), g(k) \rangle) \ \&$
 $\lambda \langle d, y \rangle \exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(y)] (\langle d, \text{mina} \rangle)$
 $= \exists x [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(g(k))] \ \&$
 $\exists x [\text{apple}(x) \ \& \ \text{large}(d)(x) \ \& \ \text{eat}(x)(\text{mina})], \text{ where } d < g(i)$
 $= \exists x \exists y \exists d [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(g(k)) \ \&$
 $\text{apple}(y) \ \& \ \text{large}(d)(y) \ \& \ \text{eat}(y)(\text{mina}) \ \& \ d < g(i)]$

Next, let's see how the topic is interpreted. Since a compared phrase that precedes the *pota*-phrase has moved from a position after the *pota*-phrase, the two positions are coindexed. When the topic is interpreted, first the meaning of the rest is abstracted

over the values for the trace, as in (50.e). That is, from (50.d), we get the meaning of a predicate, as given in (50.f). This takes the topic as the argument, as in (50.g). By this process we get the same meaning as the corresponding sentence in which the compared phrase has not moved:

- (50) e. $\llbracket [\text{Inho-nun}]_k \alpha \rrbracket = \lambda_Z \llbracket \alpha \rrbracket^{g[z/k]}(\text{inho})$
 f. $\lambda_Z \exists x \exists y \exists d [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(z) \ \& \ \text{apple}(y) \ \& \ \text{large}(d)(y) \ \& \ \text{eat}(y)(\text{mina}) \ \& \ d < g(i)]$
 (50) g. $\llbracket [\text{TP} [\text{Inho-nun}]_k [\text{TP} \text{Mina-pota} [e_k]_F [\text{VP} [\text{NP} [\text{CP} e_j [\text{VP}_2 [te_i]_F \text{khu}]]-n_i] \text{sakwa-lul}] \text{mek}]^g \rrbracket$
 $= \lambda_Z \exists x \exists y \exists d [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(z) \ \& \ \text{apple}(y) \ \& \ \text{large}(d)(y) \ \& \ \text{eat}(y)(\text{mina}) \ \& \ d < g(i)](\text{inho})$
 $= \exists x \exists y \exists d [\text{apple}(x) \ \& \ \text{large}(g(i))(x) \ \& \ \text{eat}(x)(\text{inho}) \ \& \ \text{apple}(y) \ \& \ \text{large}(d)(y) \ \& \ \text{eat}(y)(\text{mina}) \ \& \ d < g(i)]$

This analysis explains how a *pota*-phrase can be associated with the topic when the latter precedes the former.

Finally I will discuss a case where a *pota*-phrase precedes a topic. In such a case, a *pota*-phrase cannot be associated with the topic that follows. One simple idea is to assume that a topic is not focused, which is quite plausible. The meaning of a *pota*-phrase must combine with a structured meaning, and a topic does not provide a structured meaning. But one problematic possibility is that the trace of the topic can be focused at a position to the right of the *pota*-phrase (and the topic):

- (51) $[\text{John-pota} [\text{Mary-nun}]_j [\text{VP} \text{Tom-i} [e_j]_F [te_i]_F \text{cohaha}]-n\text{-ta}]$

Then we can interpret the VP as follows:

- (52) a. $\llbracket \text{cohaha} \rrbracket = \lambda_d \lambda_x \lambda_y [\text{like}(d)(x)(y)]$
 b. $\llbracket [te_i]_F \text{cohaha} \rrbracket = \langle \lambda_d \lambda_x \lambda_y [\text{like}(d)(x)(y)], g(i), \{d \in D_d \mid d < g(i)\} \rangle$
 c. $\llbracket [e_j]_F [te_i]_F \text{cohaha} \rrbracket$
 $= \langle \lambda \langle d, x \rangle \lambda y [\text{like}(d)(x)(y)], \langle g(i), g(j) \rangle, \{ \langle d, y \rangle \in D_d \times D_e \mid d < g(i) \} \rangle$
 d. $\llbracket \text{Tom-i} [e_j]_F [te_i]_F \text{cohaha} \rrbracket$
 $= \langle \lambda \langle d, x \rangle [\text{like}(d)(x)(\text{tom})], \langle g(i), g(j) \rangle, \{ \langle d, y \rangle \in D_d \times D_e \mid d < g(i) \} \rangle$

Next we need to interpret the topic, but there is no acceptable interpretation. Since the topic is not a focus-sensitive operator, the topic is supposed to be incorporated in the background of the structured meaning. But then abstracting over the possible values of e_j is not properly defined:

- (52) e. $\llbracket [\text{Mary-nun}]_j \alpha \rrbracket = \lambda z \llbracket \alpha \rrbracket^{g(z/i)}(\text{mary})$
 f. $\llbracket [\text{Mary-nun}]_j [\text{Tom-i } [e_i]_F [te_i]_F \text{ cohaha}] \rrbracket$
 $= \langle \lambda \langle d, x \rangle [\lambda z [\text{like}(d)(x)(\text{tom})](\text{mary})], \langle g(i), z \rangle, \{ \langle d, y \rangle \in \text{ALT}(g(i)) \times \text{ALT}(g(j)) \mid d \langle g(i) \rangle \}$

$g(j)$ in (52.d) is replaced with z , but it is not in the scope of the lambda operator λz . In order to put z in the scope of the lambda operator, the abstraction could apply to the whole structured meaning:

- (52) f'. $\lambda z [\langle \lambda \langle d, x \rangle [\text{like}(d)(x)(\text{tom})], \langle g(i), z \rangle, \{ \langle d, y \rangle \in \text{ALT}(g(i)) \times \text{ALT}(g(j)) \mid d \langle g(i) \rangle \}](\text{mary})$
 $= \langle \lambda \langle d, x \rangle [\text{like}(d)(x)(\text{tom})], \langle g(i), \text{mary} \rangle, \{ \langle d, y \rangle \in \text{ALT}(g(i)) \times \text{ALT}(g(j)) \mid d \langle g(i) \rangle \}$

 Focus

But one crucial problem with this interpretation is that *Mary-nun* is a topic, but that the meaning actually goes into the focus part in the structured meaning. This should be prohibited. Thus we can include the following restriction in the process of interpretation:

- (53) The reference of a topic does not go into the focus part of a structured meaning.

This is just a default assumption, which yields no cost in the grammar.

We saw that (51) leads to a crash in semantic interpretation. The only acceptable structure of (29.b) is the following:

- (54) $\llbracket [\text{John-pota } [[\text{Mary}]_j\text{-nun } [[\text{Tom-i}]_F e_j [te_i]_F \text{ cohaha}]\text{-n}]]\text{-ta} \rrbracket$.

That is, the trace of the topic cannot be focused. This sentence can be interpreted as follows:

- (55) a. $\llbracket \text{cohaha} \rrbracket = \lambda d \lambda x \lambda y [\text{like}(d)(x)(y)]$
 b. $\llbracket [te_i]_F \text{ cohaha} \rrbracket = \langle \lambda d \lambda x \lambda y [\text{like}(d)(x)(y)], g(i), \{ d \in D_d \mid d \langle g(i) \rangle \} \rangle$
 c. $\llbracket e_j [te_i]_F \text{ cohaha} \rrbracket = \langle \lambda d \lambda y [\text{like}(d)(g(j))(y)], g(i), \{ d \in D_d \mid d \langle g(i) \rangle \} \rangle$
 d. $\llbracket [\text{Tom-i}]_F e_j [te_i]_F \text{ cohaha} \rrbracket$
 $= \langle \lambda \langle d, x \rangle [\text{like}(d)(g(j))(x)], \langle g(i), \text{tom} \rangle, \{ \langle d, x \rangle \in D_d \times D_e \mid d \langle g(i) \rangle \} \rangle$
 e. $\llbracket [\text{Mary-nun}]_j \alpha \rrbracket = \lambda z \llbracket \alpha \rrbracket^{g(z/i)}(\text{mary})$
 f. $\llbracket [\text{Mary-nun}]_j [\text{Tom-i}]_F e_j [te_i]_F \text{ cohaha} \rrbracket$
 $= \langle \lambda \langle d, x \rangle [\lambda z [\text{like}(d)(z)(x)](\text{mary})], \langle g(i), \text{tom} \rangle, \{ \langle d, x \rangle \in D_d \times D_e \mid d \langle g(i) \rangle \} \rangle$
 $= \langle \lambda \langle d, x \rangle [\text{like}(d)(\text{mary})(x)], \langle g(i), \text{tom} \rangle, \{ \langle d, x \rangle \in D_d \times D_e \mid d \langle g(i) \rangle \} \rangle$

- g. $\llbracket \text{John-pota } [\text{Mary-nun}]_j [\text{Tom-i}]_F e_i \text{ te } [e_i]_F \text{ cohaha} \rrbracket$
 = $\llbracket \text{like}(g(i))(\text{mary})(\text{tom}) \rrbracket \ \& \ \llbracket \text{like}(d)(\text{mary})(\text{john}) \rrbracket$, where
 $\langle d, \text{john} \rangle \in \{ \langle d, x \rangle \in D_d \times D_e \mid d < g(i) \}$
 = $\exists d \llbracket \text{like}(g(i))(\text{mary})(\text{tom}) \ \& \ \text{like}(d)(\text{mary})(\text{john}) \ \& \ g(i) > d \rrbracket$

The trace e_i of the topic is not focused, so it is just interpreted as the internal argument without introducing a set of alternatives, as in (55.c). Instead, the nominative case-marked subject is focused. This leads to a structured meaning with a set of alternative pairs involved, as in (55.d). And the topic is normally incorporated into the background part of the structured meaning, as in (55.f). In (55.g), where the *pota*-phrase is interpreted, we get the meaning of correct comparison between Tom and John in terms of their degrees of liking Mary. This way of interpretation makes sure that if a *pota*-phrase precedes a topic, it is only associated with a non-topical compared phrase that follows.

3.3. *Pota*-phrases associated with other phrases than NPs

So far I have only discussed cases where *pota* takes a NP complement and a *pota*-phrase is associated with a NP argument. I need to consider cases where a *pota*-phrase is associated with other phrases than NPs. To interpret an adjunct PP, I need to consider event arguments, which makes the discussion a little more complex. To make the discussion simple, I will only give an explanation that is sketchy but sufficient to explain the distributions of *pota*-phrases.

When the compared phrase and the complement of *pota* are of the same phrasal type, as in (18)-(19), we can interpret them in the same way as we do when they are NPs. The meaning of (56) is derived as in (57). The only complexity comes from the semantic type of a PP. An adjunct PP is a predicate that is predicated of an event, as shown in (57.b). Thus the alternative is introduced by a variable Q which ranges over predicates of events. The *pota*-phrase is interpreted in (57.c), and (57.d) is the meaning of the sentence:

- (56) Inho-nun Seoul-eyse-pota [Busan-eyse]_F [te]_F hayngpokhay-ss-ta.
 Inho-top Seoul-in-than Busan-in more happy_V-pst-dec
 'Inho was happier in Busan than in Seoul.'
- (57) a. $\llbracket [te]_F \text{ hayngpokhay} \rrbracket^g = \langle \lambda d \lambda x \exists e [\text{happy}(e)(d)(x)], g(i), \{d' \in D_d \mid d' < g(i)\} \rangle$
 b. $\llbracket [\text{Busan-eyse}]_F [te]_F \text{ hayngpokhay} \rrbracket^g =$
 $\langle \lambda \langle d, P \rangle \lambda x \exists e [\text{happy}(e)(d)(x) \ \& \ P(e)], \langle g(i), \lambda e [\text{loc}(\text{busan}, e)] \rangle,$
 $\{ \langle d', Q \rangle \in D_d \times D_P \mid d' < g(i) \} \rangle$
 c. $\llbracket \text{Seoul-eyse-pota } [\text{Busan-eyse}]_F [te]_F \text{ hayngpokhay} \rrbracket^g =$
 $\lambda x \exists d' [\exists e [\text{happy}(e)(g(i))(x) \ \& \ \text{loc}(\text{bus.})(e)] \ \& \ \exists e' [\text{happy}(e')(d')(x) \ \& \ \text{loc}(\text{seo.})(e')]]$,
 where $\langle d', \lambda d [\text{loc}(\text{seo.})(e)] \rangle \in \{ \langle d', Q \rangle \in D_d \times D_P \mid d' < g(i) \}$
 d. $\exists d' \exists e \exists e' [\text{happy}(e)(g(i))(\text{inho}) \ \& \ \text{loc}(\text{busan})(e) \ \& \ \text{happy}(e')(d')(\text{inho}) \ \& \ \text{loc}(\text{seoul})(e') \ \& \ g(i) > d']$

Next I will discuss cases where a compared phrase has an oblique case but the *pota*-phrase takes a NP with no oblique case. In a *pota*-phrase, the standard of comparison can take an oblique case-marker optionally, as pointed out in Jhang (2001). And when the complement of *pota* takes no oblique case-marker, a *pota*-phrase can occur in various positions before the compared phrase:

- (58) a. Inho-nun Seoul-pota Busan-eyse te hayngpokhay-ss-ta.
 Inho-top Seoul-than Busan-in more happy_V-pst-dec
 'Inho earned more money in Busan than Seoul.' (Lit.)
 b. Seoul-pota Inho-nun Busan-eyse te hayngpokhay-ss-ta.

But a *pota*-phrase cannot follow the compared phrase:

- (59) a. ?*Inho-nun Busan-eyse Seoul-pota te hayngpokhay-ss-ta.
 b. ?*Busan-eyse Inho-nun Seoul-pota te hayngpokhay-ss-ta.

These can be accounted for by analyzing the *pota*-phrase as compared with a NP, not with a PP. Thus we assume that only the NP in the PP is focused:

- (58.a') Inho-nun Seoul-pota [Busan]_F-eyse [te]_F hayngpokhay-ss-ta.
 (60) a. $\llbracket \text{[Busan]}_F\text{-eyse [te]}_F \text{ hayngpokhay} \rrbracket^g =$
 $\langle \lambda \langle d, y \rangle \lambda x \exists e [\text{happy}(e)(d)(x) \ \& \ \text{loc}(y)(e)], \langle g(i), \text{busan} \rangle,$
 $\{ \langle d', z \rangle \in D_d \times D_e \mid d' < g(i) \} \rangle$
 b. $\llbracket \text{Seoul-pota [Busan]}_F\text{-eyse te}_F \text{ hayngpokhay} \rrbracket^g =$
 $\lambda x \exists d' [\exists e [\text{happy}(e)(g(i))(x) \ \& \ \text{loc}(\text{busan})(e)] \ \& \$
 $\exists e' [\text{happy}(e')(d')(x) \ \& \ \text{loc}(\text{seoul})(e')]],$
 where $\langle d', \text{seoul} \rangle \in \{ \langle d', z \rangle \in D_d \times D_e \mid d' < g(i) \}$
 c. $\llbracket \text{Inho-nun Seoul-pota [Busan]}_F\text{-eyse [te]}_F \text{ hayngpokhay} \rrbracket^g =$
 $\exists d' \exists e \exists e' [\text{happy}(e)(g(i))(\text{inho}) \ \& \ \text{loc}(\text{busan})(e) \ \& \$
 $\text{happy}(e')(d')(\text{inho}) \ \& \ \text{loc}(\text{seoul})(e') \ \& \ g(i) > d']$

(60.a) can be compared with (57.b) in that alternative NPs, not alternative PPs, are involved. The rest of the interpretations just follows.

If the *pota*-phrase with no oblique case in it follows the compared phrase with an oblique case, as in (59), it would be associated with a trace of the compared phrase. But since the complement of *pota* has no oblique case, the compared phrase would have to be a NP too. But the trace is a PP. Thus there is a type mismatch between the meaning of a NP and that of a PP, which makes the sentence odd.

4. Conclusion

In this paper, I showed that a *pota*-phrase can occur far away from a comparative predicate, and that a *pota*-phrase also occurs in various positions with respect to the compared phrase and other phrases including topics. In accounting for the observations we made, I assume that comparative constructions involve multiple foci: one focus is on *te* and the other on the compared phrase or its trace. This requires a *pota*-phrase to precede both the compared phrase (or its trace) and the comparative morpheme. In structural terms, the former must c-command the latter two. In compositional interpretation, a *pota*-phrase takes a structured meaning we get from the multiple foci and yields the meaning we want.

By assuming that the trace of a compared phrase also can be focused, we can explain how a *pota*-phrase can be compared with a topic that precedes it. The meaning specification of a *pota*-phrase in terms of focus also prohibits a *pota*-phrase from being compared with a topic that follows it. Moreover, a focus phenomenon involves a set of alternatives, and a NP cannot be associated with a PP. This also affects the ordering of a *pota*-phrase with a NP and the compared phrase with an oblique case.

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